Global Positioning System (GPS) and Geographical Information System (GIS)

Objectives
To understand how a GPS works and how it can be combined with GIS to collect features on the landscape to add information to a map.

Materials
Cell phone
ESRI Field Maps app (download from App Store)
Computer Lab
ArcGIS Online (AGOL) login with password (provided)

Procedure
1. Overview of GPS
2. Overview of GIS
3. Explore the ESRI Field Maps app and definition list
4. Students use ESRI Field Maps app on phone to collect campus data points
5. Overview of mapping in ArcGIS Online
6. Students use laptops to analyze the point data in AGOL
7. Students create their own map using data points collected earlier
8. Explanation of how students can create their own ArcGIS Online account

The Unmanned Aerial Vehicle (UAV) aka Drone

Objectives
To understand the uses of an UAV as related to GPS and GIS, learn the basics of flying, and some basic rules for drone use.

Materials
UAV (Drone)
Drone course

Procedure
1. Overview of how drones are used with GPS and GIS in the industry
2. Overview of flight basics with drones
3. Overview of rules for drone use (e.g., airspace/property rights, weather)
4. Overview of flight as a team (e.g., responsibilities, communication)
5. Students learn to fly drones with basic maneuvers (e.g., forward, backward, side to side, up, down)
6. Students fly obstacle course using flight rubric for scoring (following page)

Flight Rubric

Name _____

Knowledge/Skills	Points
Team communication	/10
Launch	/1
Flight – land on target (points awarded based on rings on target)	/5
Flight – fly through hoop	/3
Flight – successfully fly three obstacles in a row	/10
Controlled landing	/4
Hit the wall or ceiling OR stuck in net (each time)	/-1
Hit drone course obstacle (each time)	/-1

Teammate Name _____

Knowledge/Skills	Points
Team communication	/10
Launch	/1
Flight – land on target (points awarded based on rings on target)	/5
Flight – fly through hoop	/3
Flight – successfully fly three obstacles in a row	/10
Controlled landing	/4
Hit the wall or ceiling OR stuck in net (each time)	/-1
Hit drone course obstacle (each time)	/-1